Answers learning task 16

2026 – 2027

Exercise 9.42: A melting point test of $n = 15$ samples of a binder used in manufacturing a rocket propellant resulted in $\overline{x} = 68°$C. Assume that the melting point is normally distributed with $\sigma = 0.83°$C.

(a) Test $H_0 : \mu = 68.3$ versus $H_1 : \mu \neq 68.3$ using $\alpha = 0.01$.

(b) What is the p-value for this test?

(c) What is the β-error if the true mean is $\mu = 65.5$?

Answer

(a) Notice that the mean melting point of 15 samples is normally distributed: $\overline{X} \sim N(\mu = ?, \sigma = \frac{0.83}{\sqrt{15}})$. We perform a two-sided hypothesis test with $H_0 : \mu = 68.3$ versus $H_1 : \mu \neq 68.3$. Under the null hypothesis, we assume that $\mu = 68.3$ and check how probable is the sample mean value $\overline{x} = 68$. The critical region is of the form $Z = (-\infty, C_L] \cup [C_R, \infty)$, where

$$C_L = \text{InvNorm}(\text{area} = \alpha/2 = 0.005, \mu = 68.3, \sigma = \frac{0.83}{\sqrt{15}}) \approx 67.7480.$$
$$C_R = \text{InvNorm}(\text{area} = 1 - \alpha/2 = 0.995, \mu = 68.3, \sigma = \frac{0.83}{\sqrt{15}}) \approx 68.8520.$$

The sample mean $\overline{x} = 68$ does not lie in the critical region, hence we fail to reject H_0. Based on this sample, there is not evidence to reject the claim that the true mean is $\mu = 68.3$.

Alternative: we can also consider the z-score of the sample mean and compute the critical region in terms of the standard normal distribution. In that case, the z-score of the sample mean equals $z = \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{68 - 68.3}{0.83/\sqrt{15}} \approx -1.3999$. The critical region is of the form $Z = (-\infty, C_L] \cup [C_R, \infty)$, where

$$C_L = \text{InvNorm}(\text{area} = \alpha/2 = 0.005, \mu = 0, \sigma = 1) \approx -2.5758.$$
$$C_R = \text{InvNorm}(\text{area} = 1 - \alpha/2 = 0.995, \mu = 0, \sigma = 1) \approx 2.5758.$$

The z-score of the sample mean $z = -1.3999$ does not lie in the critical region, hence we fail to reject H_0. Based on this sample, there is not evidence to reject the claim that the true mean is $\mu = 68.3$.

(b) First, note that the sample mean $\overline{x} = 68$ is smaller than the assumed null mean $\mu_0 = 68.3$. The p-value (the probability that we get an observed sample mean

further from 68.3 than $\mu_0 = 68$) can therefore be computed as follows:

$$p = 2 \cdot P(\overline{X} \leq \overline{x}) = \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = 68, \mu = 68.3, \sigma = \frac{0.83}{\sqrt{15}}) \approx 0.1616.$$

(c) The β-error (probability of not rejecting H_0 when H_1 is true) can be computed by the probability of observing a sample mean in the acceptance region, given the true mean $\mu = 65.5$:

$$\begin{aligned}\beta = P(H_0 \mid H_1) &= P(C_L \leq \overline{X} \leq C_R \mid \mu = 65.5) \\ &= P(67.7480 \leq \overline{X} \leq 68.8520 \mid \mu = 65.5) \\ &= \text{normalcdf}(\text{lower} = 67.7480, \text{upper} = 68.8520, \mu = 65.5, \sigma = \frac{0.83}{\sqrt{15}}) \\ &\approx 4.9268 \cdot 10^{-26}.\end{aligned}$$

Exercise 9.47: Medical researchers have developed a new artificial heart constructed primarily of titanium and plastic. The heart will last and operate almost indefinitely once it is implanted in the patient's body, but the battery pack needs to be recharged about every four hours. A random sample of 50 battery packs is selected and subjected to a life test. The average life of these batteries is 4.05 h. Assume that battery life is normally distributed with standard deviation $\sigma = 0.2$ h.

(a) Is there evidence to support the claim that mean battery life exceeds 4 h? Use $\alpha = 0.05$.

(b) What is the p-value for the test in part (a)?

(c) Compute the power of the test if the true mean battery life is 4.5 h.

Answer

(a) Notice that the mean battery life of 50 battery packs is normally distributed: $\overline{X} \sim N(\mu = ?, \sigma = \frac{0.2}{\sqrt{50}})$. We perform a right-sided hypothesis test with $H_0 : \mu = 4$ versus $H_1 : \mu > 4$. Under the null hypothesis, we assume that $\mu = 4$ and check how probable is the sample mean value $\overline{x} = 4.05$. The critical region is of the form $Z = [C_R, \infty)$, where

$$C_R = \text{InvNorm}(\text{area} = 1 - \alpha = 0.95, \mu = 4, \sigma = \frac{0.2}{\sqrt{50}}) \approx 4.0465.$$

The sample mean $\overline{x} = 4.05$ lies inside the critical region, hence we reject H_0. Based on this sample, there is enough evidence to support the claim that the mean battery life indeed exceeds four hours.

(b) First, note that we consider a right-sided hypothesis test, therefore the p-value (the probability that we will observe a sample mean even larger than 4.05) can be computed as follows:

$$p = P(\overline{X} \geq \overline{x}) = \text{normalcdf}(\text{lower} = 4.05, \text{upper} = 10^{99}, \mu = 4, \sigma = \frac{0.2}{\sqrt{50}}) \approx 0.0385.$$

(c) The power of a hypothesis test is $1 - \beta$, where β is the type II error probability (probability of not rejecting H_0 when H_1 is true). With other words, the power

is the probability of observing a sample mean in the acceptance region, given the true (alternative) mean $\mu = 4.5$:

$$\begin{aligned} 1-\beta = P(H_1 \mid H_1) &= P(\overline{X} \geq C_R \mid \mu = 4.5) \\ &= P(\overline{X} \geq 4.0465 \mid \mu = 4.5) \\ &= \text{normalcdf}(\text{lower} = 4.0465, \text{upper} = 10^{99}, \mu = 4.5, \sigma = \frac{0.2}{\sqrt{50}}) \\ &= 1. \end{aligned}$$

Exercise 9.51: For the hypothesis test $H_0 : \mu = 7$ against $H_1 : \mu \neq 7$ with variance unknown and $n = 25$, approximate the p-value for each of the following test statistics.

(a) $t_0 = 2.05$ (b) $t_0 = -1.84$ (c) $t_0 = 0.4$

Answer

In all three cases, we consider a two-sided hypothesis test for the mean μ of a normal distribution (with unknown σ). Therefore, we need to resort to the t-distribution.

The test statistic we are working with is

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}},$$

where $\overline{X}$ is the sample mean and S is the sample standard deviation. This test statistic follows a t-distribution with $\text{df} = n - 1 = 24$ degrees of freedom.

As we consider two-sided tests, the p-values of given t_0-values can be computed as follows:

$$p = \begin{cases} 2 \cdot P(T \geq t_0) = \text{tcdf}(\text{lower} = t_0, \text{upper} = 10^{99}, \text{df} = 24), & \text{if } t_0 \geq 0 \\ 2 \cdot P(T < t_0) = \text{tcdf}(\text{lower} = -10^{99}, \text{upper} = t_0, \text{df} = 24), & \text{otherwise.} \end{cases}$$

(a) Filling in $t_0 = 2.05$ gives

$$p = 2 \cdot P(T \geq 2.05) = \text{tcdf}(\text{lower} = 2.05, \text{upper} = 10^{99}, \text{df} = 24) \approx 0.0514.$$

(b) Filling in $t_0 = -1.84$ gives

$$p = 2 \cdot P(T < -1.84) = \text{tcdf}(\text{lower} = -10^{99}, \text{upper} = -1.84, \text{df} = 24) \approx 0.0782.$$

(c) Filling in $t_0 = 0.4$ gives

$$p = 2 \cdot P(T \geq 0.4) = \text{tcdf}(\text{lower} = 0.4, \text{upper} = 10^{99}, \text{df} = 24) \approx 0.6927.$$

Exercise 9.59: A 1992 article in the *Journal of the American Medical Association* ("A Critical Appraisal of 98.6 Degrees F, the Upper Limit of the Normal Body Temperature, and Other Legacies of Carl Reinhold August Wunderlich") reported body temperature, gender, and heart rate for a number of subjects. The body temperatures for 25 female subjects follow: 97.8, 97.2, 97.4, 97.6, 97.8, 97.9, 98.0, 98.0, 98.0, 98.1, 98.2, 98.3, 98.3, 98.4, 98.4, 98.4, 98.5, 98.6, 98.6, 98.7, 98.8, 98.8, 98.9, 98.9, and 99.0.

(a) Test the hypothesis $H_0 : \mu = 98.6$ versus $H_1 : \mu \neq 98.6$, using $\alpha = 0.05$. Find the p-value.

(e) Explain how the question in part (a) could be answered by constructing a two-sided confidence interval on the mean female body temperature.

Answer

(a) We consider a two-sided t-test for the mean μ of a normal distribution, as the standard deviation σ is unknown. First, we need to compute the sample mean $\overline{x}$ and the sample standard deviation s of the given sample:

$$\overline{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{97.8 + 97.2 + \ldots + 99.0}{25} \approx 98.2640.$$

$$\begin{aligned} s^2 &= \frac{(x_1 - \overline{x})^2 + (x_2 - \overline{x})^2 + \ldots + (x_n - \overline{x})^2}{n-1} \\ &= \frac{(97.8 - 98.2640)^2 + (97.2 - 98.2640)^2 + \ldots + (99.0 - 98.2640)^2}{25 - 1} \\ &\approx 0.2324. \end{aligned}$$

$$s = \sqrt{s^2} = \sqrt{0.2324} \approx 0.4821.$$

The test statistic for the t-test is then computed as

$$t_0 = \frac{\overline{x} - \mu}{s/\sqrt{n}} = \frac{98.2640 - 98.6}{0.4821/\sqrt{25}} \approx -3.4849.$$

The critical region ($\alpha = 0.05$, two-sided t-test) is of the form $Z = (-\infty, C_L] \cup [C_R, \infty)$, where:

$$\begin{aligned} C_L &= \mathrm{InvT}(\text{area} = \alpha/2 = 0.025, \text{df} = 24) \approx -2.0639, \\ C_R &= \mathrm{InvT}(\text{area} = 1 - \alpha/2 = 0.975, \text{df} = 24) \approx 2.0639. \end{aligned}$$

The observed test statistic t_0 lies inside the critical region, therefore we reject H_0.

Alternatively, we could have computed the p-value corresponding to t_0:

$$\begin{aligned} p = 2 \cdot P(T \leq -3.4849) \quad &= 2 \cdot \mathrm{tcdf}(\text{lower} = -10^{99}, \text{upper} = -3.4849, \text{df} = 24) \\ &\approx 0.0019. \end{aligned}$$

As $p < \alpha$, we again decide to reject H_0. There is enough evidence to assume that the mean female body temperature significantly differs from $98.6°F$.

(e) A two-sided confidence interval for the mean μ can be computed as follows:

$$[\overline{x} - t_{\alpha/2;n-1} \cdot \frac{s}{\sqrt{n}}, \overline{x} + t_{\alpha/2;n-1} \cdot \frac{s}{\sqrt{n}}]$$

Here, $\overline{x} = 98.2640$, $s = 0.4821$, $n = 25$ and

$$t_{\alpha/2,n-1} = \mathrm{InvT}(\text{area} = 1 - \alpha/2 = 0.975, \text{df} = n - 1 = 24) \approx 2.0639.$$

Filling in this information in the formula gives an interval $[98.0650, 98.4630]$. Notice that 98.6 does not lie inside this confidence interval, hence it is very unlikely that 98.6 is the true value of the mean μ. Therefore, we can again decide to reject H_0.

Exercise 9.60: Cloud seeding has been studied for many decades as a weather modification procedure (for an interesting study of this subject, see the article in *Technometrics*, "A Bayesian Analysis of a Multiplicative Treatment Effect in Weather Modification," Vol. 17, pp. 161–166). The rainfall in cubic-meters from 20 clouds that were selected at random and seeded with silver nitrate follows: 22.2×10^3, 37.9×10^3, 24.4×10^3, 33.4×10^3, 27.5×10^3, 23.2×10^3, 39.2×10^3, 28.9×10^3, 26.1×10^3, 34.4×10^3, 39.3×10^3, 33.4×10^3, 30.8×10^3, 30.4×10^3, 33.2×10^3, 26.9×10^3, 36.2×10^3, 42.9×10^3, 32.9×10^3, and 38.9×10^3.

(a) Can you support a claim that the mean rainfall from seeded clouds exceeds $30 \times 10^3\,\mathrm{m}^3$? Use $\alpha = 0.01$. Find the p-value.

(e) Explain how the question in part (a) could be answered by constructing a one-sided confidence bound on the mean rainfall in cubic metres.

Answer

(a) We consider a right-sided t-test for the mean μ of a normal distribution, as the standard deviation σ is unknown. First, we need to compute the sample mean $\overline{x}$ and the sample standard deviation s of the given sample:

$$\overline{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{22.2 + 37.9 + \ldots + 38.9}{20} \approx 32.1050.$$

$$\begin{aligned} s^2 &= \frac{(x_1 - \overline{x})^2 + (x_2 - \overline{x})^2 + \ldots + (x_n - \overline{x})^2}{n-1} \\ &= \frac{(22.2 - 32.1050)^2 + (37.9 - 32.1050)^2 + \ldots + (38.9 - 32.1050)^2}{20 - 1} \\ &\approx 34.8226. \end{aligned}$$

$$s = \sqrt{s^2} = \sqrt{34.8226} \approx 5.9011.$$

The test statistic for the t-test is then computed as

$$t_0 = \frac{\overline{x} - \mu}{s/\sqrt{n}} = \frac{32.1050 - 30}{5.9011/\sqrt{20}} \approx 1.5953.$$

The critical region ($\alpha = 0.01$, right-sided t-test) is of the form $Z = [C_R, \infty)$, where:

$$C_R = \mathrm{InvT}(\text{area} = 1 - \alpha = 0.99, \text{df} = 19) \approx 2.5395.$$

The observed test statistic $t_0 \approx 1.5953$ does not lie inside the critical region, therefore we fail to reject H_0.

Alternatively, we could have computed the p-value corresponding to t_0:

$$p = P(T \geq 1.5953) = \mathrm{normalcdf}(\text{lower} = 1.5953, \text{upper} = 10^{99}, \text{df} = 19) \approx 0.0636.$$

As $p > \alpha = 0.05$, we again fail to reject H_0. There is not enough evidence to assume that the mean rainfall from a random cloud significantly exceeds $30 \times 10^3\,\mathrm{m}^3$.

(e) An alternative way of answering part (a) is through a one-sided confidence interval for the mean μ. As we consider a right-sided hypothesis test, this means that the

confidence interval is of the form $[\overline{x} - t_{\alpha,n-1} \cdot s/\sqrt{n}, \infty)$, In this case, $n = 20$ and $\alpha = 0.01$, so

$$t_{\alpha,n-1} = \text{InvT}(\text{area} = 1 - \alpha = 0.99, \text{df} = n - 1 = 19) \approx 2.5395.$$

The one-sided confidence interval is therefore equal to

$$[\overline{x} - t_{\alpha,n-1} \cdot s/\sqrt{n}, \infty) = [32.1050 - 2.5395 \cdot 5.9011/\sqrt{20}, \infty) = [28.7540, \infty).$$

Notice that $\mu = 30$ is inside this confidence interval, hence we fail to reject H_0. There is not enough evidence to assume that the mean rainfall from a random cloud significantly exceeds $30 \times 10^3\,\text{m}^3$.

Exercise 9.126: Consider the following computer output.

One-Sample Z:
Test of $\mu = 26$ vs $\mu > 26$
The assumed standard deviation $= 1.7$

Variable	n	Mean	StDev	SE Mean	Z	p
X	?	26.541	2.032	0.401	?	?

(a) Fill in the missing information.

(b) Is this a one-sided or a two-sided test?

(c) What are your conclusions if $\alpha = 0.05$?

(d) Find a $95\,\%$ two-sided CI on the mean.

Answer

(a) From the given information, we can derive the following missing descriptive statistics:

- Notice that the standard error for the mean (SE Mean) follows from the formula
$$\text{SE Mean} = \frac{\sigma}{\sqrt{n}},$$
or alternatively $n = \left(\frac{\sigma}{\text{SE Mean}}\right)^2 = \left(\frac{1.7}{0.401}\right) = 18$.
- The z-score corresponding to a sample with sample mean 26.541, $\sigma = 1.7$ and $n = 18$ is equal to:
$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{26.541 - 26}{1.7/\sqrt{18}} \approx 1.3502.$$
- The p-value corresponding to this sample is equal to
$$\begin{aligned} p &= P(\overline{X} \geq \overline{x} = 26.541) \\ &= \text{normalcdf}(\text{lower} = 26.541, \text{upper} = 10^{99}, \mu = 26, \sigma = 1.7/\sqrt{18}) \\ &\approx 0.0885. \end{aligned}$$

(b) This is a one-sided hypothesis test (specifically, a right-sided test), as the alternative hypothesis is of the form "$>$".

(c) If the significance level $\alpha = 0.05$, we have that $p = 0.0885 > \alpha = 0.05$, hence we fail to reject H_0. There is not enough evidence to reject the claim that the true mean $\mu = 26$.

(d) A two-sided $95\,\%$ ($\alpha = 0.05$) confidence interval for the mean μ can be computed as follows:

$$[\overline{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \overline{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}]$$

Here, $\overline{x} = 26.541$, $\sigma = 1.7$, $n = 18$ and

$$z_{\alpha/2} = \text{InvNorm}(\text{area} = 1 - \alpha/2 = 0.975, \mu = 0, \sigma = 1) \approx 1.9600.$$

Filling in this information in the formula gives an interval

$$\begin{aligned}
&[\overline{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \overline{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}] \\
&= [26.541 - 1.9600 \cdot \frac{1.7}{\sqrt{18}}, 26.541 + 1.9600 \cdot \frac{1.7}{\sqrt{18}}] \\
&= [25.7556, 27.3264].
\end{aligned}$$

Exercise 9.127: Consider the following computer output.

One-Sample T:
Test of $\mu = 100$ vs $\mu \neq 100$

Variable	n	Mean	StDev	SE Mean	$95\,\%$ CI	t	p
X	16	98.33	4.61	?	(?,?)	?	?

(a) How many degrees of freedom are there on the t-statistic?

(b) Fill in the missing information. You may use bounds on the p-value.

(c) What are your conclusions if $\alpha = 0.01$?

(d) What are your conclusions if the hypothesis is $H_0 : \mu = 100$ versus $H_0 : \mu > 100$?

Answer

(a) The number of degrees of freedom equals $\text{df} = n - 1$, where $n = 16$ is the sample size. Therefore, $\text{df} = 15$.

(b) From the given information, we can derive the following missing descriptive statistics:

- Notice that the standard error for the mean (SE Mean) follows from the formula
$$\text{SE Mean} = \frac{\text{StDev}}{\sqrt{n}} = \frac{4.61}{\sqrt{16}} \approx 1.1525.$$
- A two-sided $95\,\%$ ($\alpha = 0.05$) confidence interval for the mean μ can be computed as follows:
$$[\overline{x} - t_{\alpha/2,n-1} \cdot \frac{\sigma}{\sqrt{n}}, \overline{x} + t_{\alpha/2,n-1} \cdot \frac{\sigma}{\sqrt{n}}]$$

Here, $\overline{x} = 98.33$, $s = 4.61$, $n = 16$ and

$$t_{\alpha/2,n-1} = \text{InvT}(\text{area} = 1 - \alpha/2 = 0.975, \text{df} = 15) \approx 2.1314.$$

Filling in this information in the formula gives an interval

$$\begin{aligned}
&[\overline{x} - t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}, \overline{x} + t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}] \\
&= [98.33 - 2.1314 \cdot \frac{4.61}{\sqrt{16}}, 98.33 + 2.1314 \cdot \frac{4.61}{\sqrt{16}}] \\
&= [95.8735, 100.7865].
\end{aligned}$$

- The t-score corresponding to a sample with $\overline{x} = 98.33$, $s = 4.61$, $\mu = 100$ and $n = 16$ is equal to:

$$t = \frac{\overline{x} - \mu}{s/\sqrt{n}} = \frac{98.33 - 100}{4.61/\sqrt{16}} \approx -1.4490.$$

- The p-value corresponding to this sample is equal to

$$p = 2 \cdot P(T \leq t) = 2 \cdot \text{tcdf}(\text{lower} = -10^{99}, \text{upper} = -1.4490, \text{df} = 15) \approx 0.1679.$$

(c) If the significance level $\alpha = 0.01$, we have that $p = 0.1679 > \alpha = 0.01$, hence we fail to reject H_0. There is not enough evidence to reject the claim that the true mean $\mu = 100$.

(d) Suppose now we consider a right-sided hypothesis test. In that case, the critical region ($\alpha = 0.01$, right-sided t-test) is of the form $Z = [C_R, \infty)$, where:

$$C_R = \text{InvT}(\text{area} = 1 - \alpha = 0.99, \text{df} = 15) \approx 2.6025.$$

The observed test statistic $t_0 \approx -1.4490$ does not lie inside the critical region, therefore we fail to reject H_0.

Alternatively, we could have computed the p-value corresponding to t_0:

$$p = P(T \geq -1.4490) = \text{tcdf}(\text{lower} = -1.4490, \text{upper} = 10^{99}, \text{df} = 15) \approx 0.9160.$$

As $p > \alpha = 0.01$, we again fail to reject H_0. There is not enough evidence to assume that the true mean is significantly larger than 100.